ClipSAT : Egyptian Scholastic Test: EST II Mathematics Level 2

Unit 3.01: Polynomial, Rational, Exponential & Logarithmic Functions

Name: _____

Q1. The graph of the function y = fx is shown below. What is the value of f1 ?

[ ] A: 0
[ ] B: 2
[ ] C: -1
[ ] D: 3

Q2. The graph of the function y = fx = ax^2 + bx + c is shown below. The graph passes through the points -1, 5 , 1, 1 , and 3, 5 . Determine the value of b .

[ ] A: 0
[ ] B: -2
[ ] C: 2
[ ] D: 1

Q3. Consider the functions fx = x^2 - 6x + 5 and gx = -x + 5 . The graphs of these two functions intersect at two points. Let these points be A and B . A third function, hx , is defined as the difference between the x-coordinates of these intersection points, i.e., hx = |x_A - x_B| . What is the value of hx ?

[ ] A: sqrt{16}
[ ] B: sqrt{25}
[ ] C: sqrt{36}
[ ] D: sqrt{49}

Q4. Consider the function fx = x^2 - 4x + 3 . What are the coordinates of the vertex of the parabola represented by this function?

[ ] A: 2, -1
[ ] B: 2, 1
[ ] C: 1, 0
[ ] D: 3, 0

Q5. The graph of the quadratic function fx = x^2 + bx + c is shown below. The graph passes through the point -2, 5 and has its vertex at 1, -4 . Determine the value of b .

[ ] A: 1
[ ] B: -2
[ ] C: -1
[ ] D: 2

Q6. The graph of the function fx = -x^2 + 4x - 3 is shown below. Determine the y-coordinate of the vertex of the parabola.

[ ] A: 1
[ ] B: 2
[ ] C: 3
[ ] D: 4

Q7. The graph of the quadratic function fx = ax^2 + bx + c is shown below. The vertex of the parabola is at 2, -5 and the y-intercept is at 0, -1 . Determine the value of a .

[ ] A: 1
[ ] B: -1
[ ] C: 2
[ ] D: -2

Q8. Consider the graph of the function fx shown below. What is the value of f0 ?

[ ] A: -1
[ ] B: 1
[ ] C: 0
[ ] D: 2